

Policy Insights

The Shifting Dynamics of Critical Minerals Trade

As demand surges, governments turn to trade policy to secure critical mineral supplies

HIGHLIGHTS

- 1 **Demand for critical energy transition minerals (CETMs) is rising rapidly** due to the expansion of clean energy technologies, electrification, and digitalization.
- 2 **Supply of CETMs is concentrated in a few countries in key segments of the value chain**, especially in processing and refining.
- 3 **Trade policy is increasingly being used to secure access to and participate in CETM supply chains**, driven by concerns over supply security, industrial competitiveness, and resilience.
- 4 **New international frameworks are emerging**, with trade policy instruments playing a central role in the reconfiguration of CETM supply chains.
- 5 **Stronger international cooperation and greater policy coherence are essential** to ensure that trade supports both the development needs of critical-mineral-rich developing countries and the global transition to a low-carbon, digital economy.



Clean energy transition fuels unprecedented demand for critical minerals

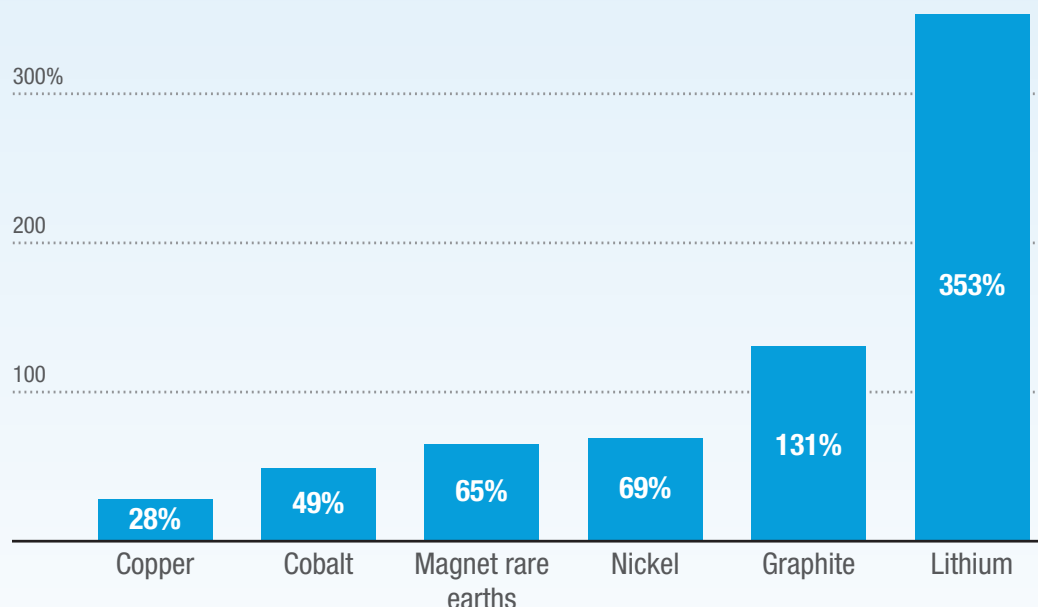
Critical energy transition minerals (CETMs)¹, including lithium, cobalt, nickel, copper, and rare earth elements, have become indispensable to the global transition toward low-carbon energy systems underpinning renewable energy technologies, battery storage and electric mobility. Other growth sectors reinforce and amplify this trend: digitalization, including data centres, semiconductors, and consumer electronics, together with defence, aerospace, and broader industrial electrification, draw on many of the same minerals. The result is sustained demand pressure.

The International Energy Agency projects strong growth in CETM demand from 2024 to 2040, with lithium and graphite set to see the strongest growth (Figure 1).

¹ As there is no single definition of critical minerals, each country determines its own based on national priorities. In this report, critical energy transition minerals (CETMs), also referred to as critical minerals for short, are defined according to UNCTAD's list of minerals required for the energy transition (see <https://sdgpulse.unctad.org/critical-minerals/index.html>).



Figure 1
Critical mineral demand is set to surge by 2040
Projected growth in demand, 2024–2040

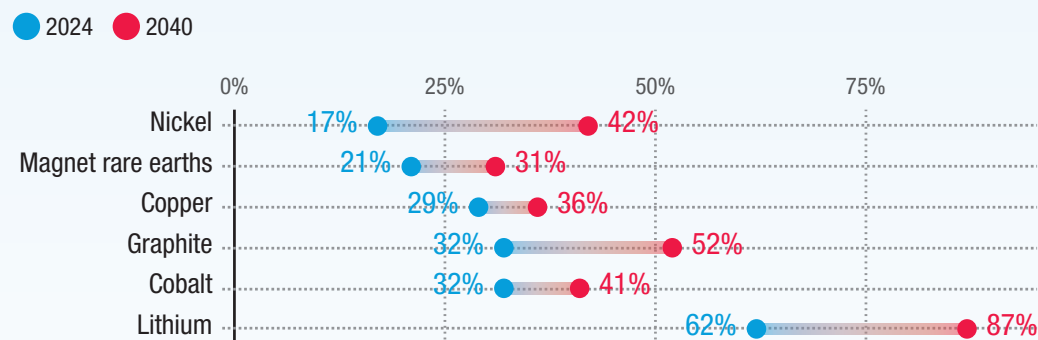


Source: UN Trade and Development (UNCTAD) calculations based on data from the International Energy Agency (IEA), Critical Minerals Outlook 2025.

Note: Based on the IEA's Stated Policies Scenario (STEPS).

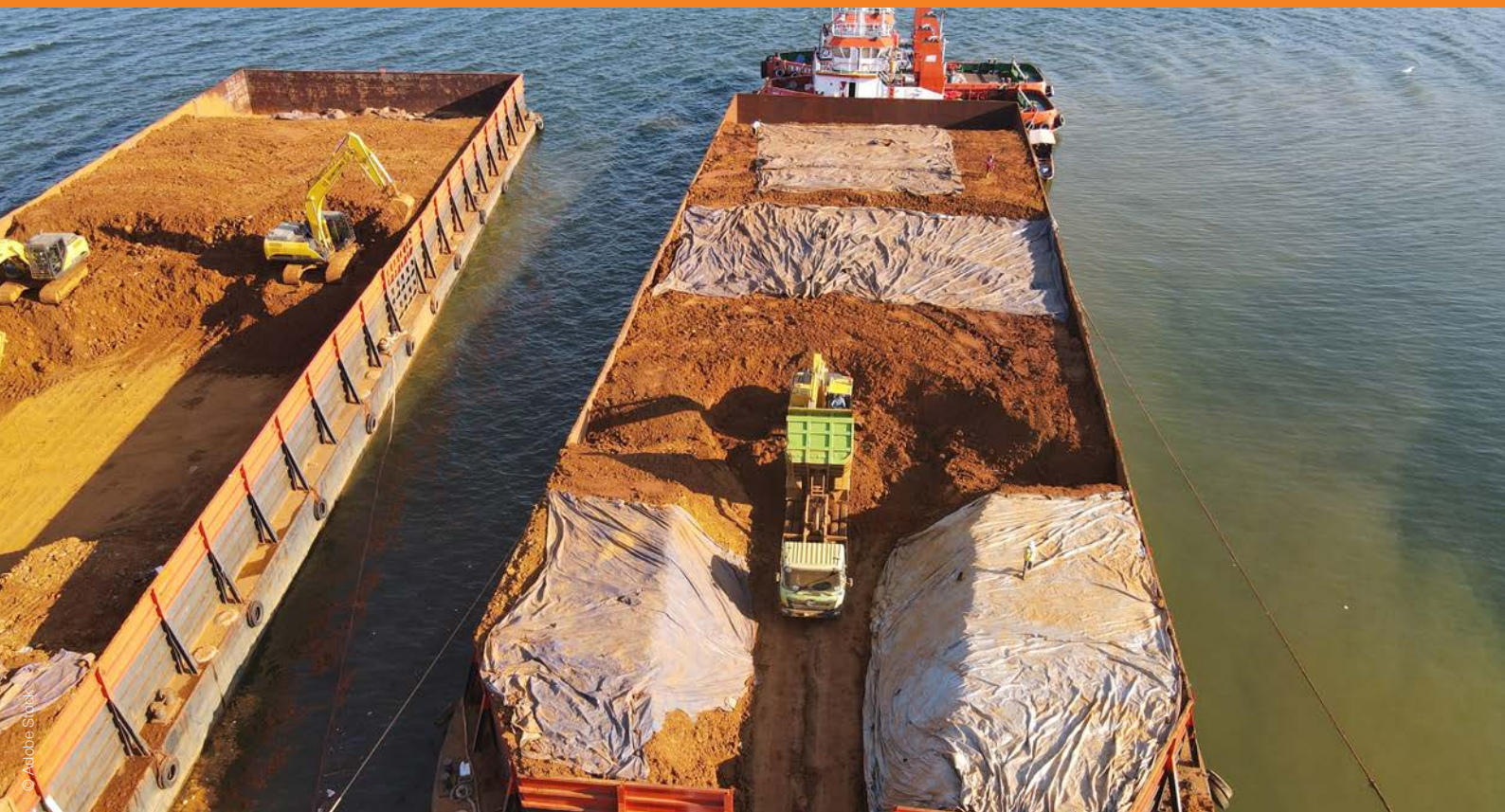
However, demand growth varies significantly across sectors. Clean technologies are expected to take a much larger share of total demand. Between 2024 and 2040, the clean technologies share of nickel demand is projected to increase from 17 per cent to 42 per cent, and of magnet rare earth elements from 21 per cent to 31 per cent (Figure 2).

Figure 2
Clean technologies are claiming a growing share of critical mineral demand



Source: UN Trade and Development (UNCTAD) calculations based on data from the International Energy Agency (IEA), Critical Minerals Outlook 2025.

Note: Based on the IEA's Stated Policies Scenario (STEPS).



Supply chains for critical minerals are highly concentrated

While each CETM faces its own individual market conditions, its supply is generally characterized by high geographical concentration at every stage of the value chain, from reserves through mining to processing and refining, with a small number of countries dominating global output.

Reserves of many key minerals including lithium, cobalt, rare earth elements, and nickel are unevenly distributed, with a few countries accounting for a dominant share (Figure 3). This concentration carries over into mining, where a small number of countries and firms dominate extraction. For example, in 2025:

The Democratic Republic of the Congo accounted for 50 per cent of global cobalt reserves and 74 per cent of global cobalt mine production.²

China accounted for 69 per cent of rare earth and 78 per cent of natural graphite mine production.³

Indonesia accounted for 67 per cent of global nickel mine production.⁴

Australia, Chile and China, together generated 72 per cent of global lithium mine production.⁵

² United States Geological Survey (USGS), Mineral Commodity Summaries 2026.

³ Ibid

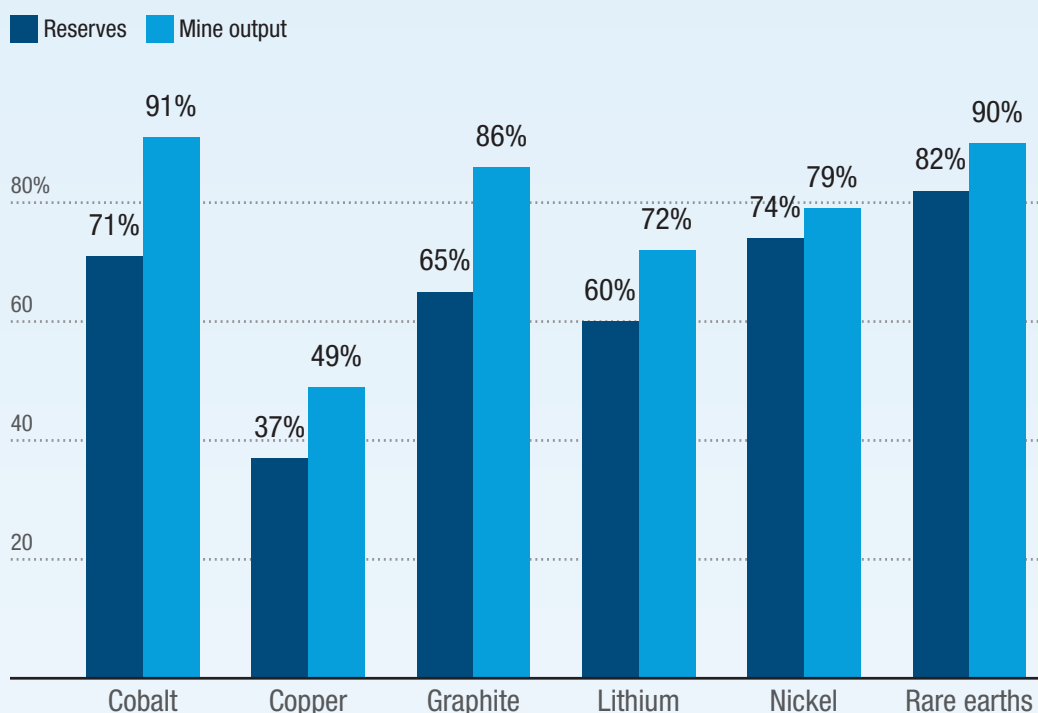
⁴ Ibid

⁵ Ibid



**Figure 3****Supply of critical minerals is highly concentrated**

Combined share of the top three countries, reserves and mine output



Source: UN Trade and Development (UNCTAD) calculations based on data from United States Geological Survey (USGS), Mineral Commodity Summaries 2026.

Mining is capital-intensive and characterized by long lead times, limiting short-term supply responsiveness and leaving concentrated supply chains exposed to geopolitical risks, governance challenges, and environmental and social pressures.

Refining and other downstream stages are even more concentrated, creating critical bottlenecks in CETM supply chains. For instance, China dominates CETM refining, including rare earths, lithium, and cobalt, while Indonesia leads in nickel refining, accounting for 43 per cent of global capacity.⁶

Refining requires substantial long-term capital investment, along with access to advanced technologies, specialized infrastructure, significant energy inputs, and economies of scale to be cost-competitive, creating high barriers to entry for new players. As a result, value addition activities are highly centralized, shaping trade patterns and raising concerns about supply chain resilience and the distribution of benefits. While efforts to diversify processing capacity, promote recycling, and strengthen domestic value chains are growing throughout different countries and regions, these transitions are long-term, require significant investment, and depend on coordinated policy support.

⁶ IEA, Critical Minerals Outlook 2025.



In addition, the main manufacturing sectors that rely on CETMs are concentrated in a few rapidly growing, technology- and energy-intensive industries. These include clean energy generation and storage, transportation, power transmission and electrical infrastructure, electronics and digital technologies (including semiconductors), and defence and aerospace. These sectors involve high value addition, technological complexity, and strong linkages to manufacturing ecosystems, which explains why countries with processing and industrial capacity can capture a substantial share of CETM-related economic benefits.

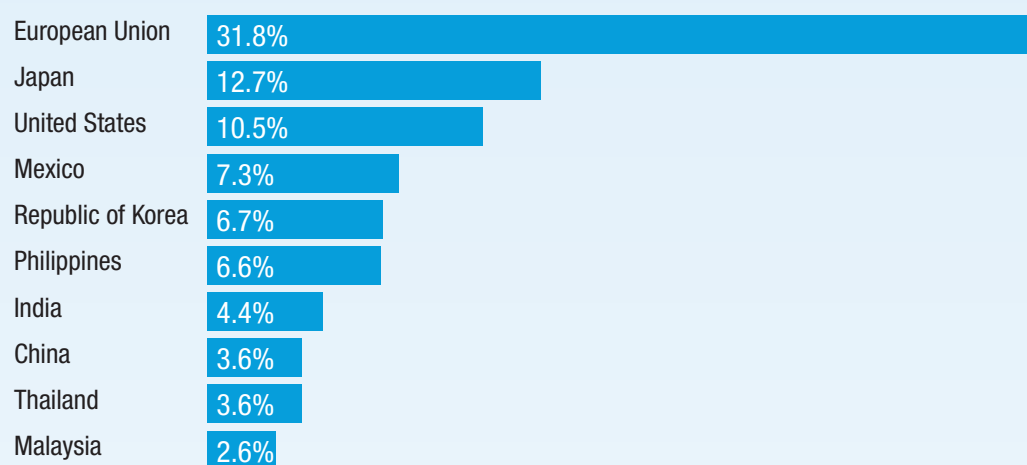
High-performance permanent magnets incorporating rare-earth elements exemplify this dynamic. They are indispensable inputs in advanced manufacturing, including electric vehicle traction motors, wind turbine generators, and defense actuators and electronics systems. Imports of rare-earth permanent magnets are highly concentrated in industrialized economies. In 2024, three importers - the European Union, Japan, and the United States - accounted for 55 per cent of global imports (Figure 4). These economies use imported magnets to produce high-value finished goods, underscoring that most CETM value is captured not at extraction, but in processing and advanced manufacturing.



Figure 4

Advanced economies dominate permanent-magnet imports

Share of global reported imports of rare-earth permanent magnets, 2024



Source: UN Trade and Development (UNCTAD) calculation based on data from COMTRADE.

Note: Calculation based on reported imports under HS code 850511.



Trade measures are increasingly reshaping critical minerals supply chains

International trade plays a central role in ensuring the availability of CETMs and rare earth elements. Cross-border trade in ores, concentrates, refined materials, and downstream components enables access to geographically dispersed stages of production across complex global value chains, particularly in high-technology sectors. As a result, most economies depend on imports at one or more stages of the value chain to meet their CETM needs. This deep interdependence heightens exposure to supply disruptions and underscores the importance of diversifying supply sources, enhancing market transparency, and strengthening international cooperation to ensure an open, stable, secure, transparent, rules-based and sustainable CETM supply.

Governments are progressively aligning CETM-related trade policies with broader strategic objectives that extend beyond cost efficiency, notably supply chain resilience, national security, and industrial competitiveness. Trade policy measures affecting CETM supply chains already span a wide range of instruments, including tariffs, quantitative restrictions, export duties, licensing regimes, technical regulations, investment measures, subsidies, trade remedies, or security-related controls, among others. Operating across multiple WTO disciplines, these instruments play a central role in shaping access to critical minerals, influencing the distribution of value added, and enhancing -or constraining- the resilience of global supply chains.



Export measures in CETM markets are on the rise

Amid rising demand for critical energy transition minerals, a number of critical-mineral-rich developing countries are adopting strategies to strengthen their position in global supply chains, aimed at capturing greater domestic value. In this context, numerous critical-mineral-rich countries have started to introduce a range of trade-restrictive measures aimed at reducing strategic dependencies and reinforcing domestic capabilities to benefit from downstream value addition, industrial development, and job creation, offering a unique opportunity to move up the value chain and reduce long-standing dependence on commodity exports.

Since 2020, there has been a surge in export-restrictive measures applied by critical-mineral-rich countries, which fall into three main types:

Licensing requirements – meant to strengthen the monitoring and control of export activities and safeguard national security

Export taxes – applied to generate additional fiscal revenues and support domestic processing

Export bans – used to encourage local processing and the development of value added activities

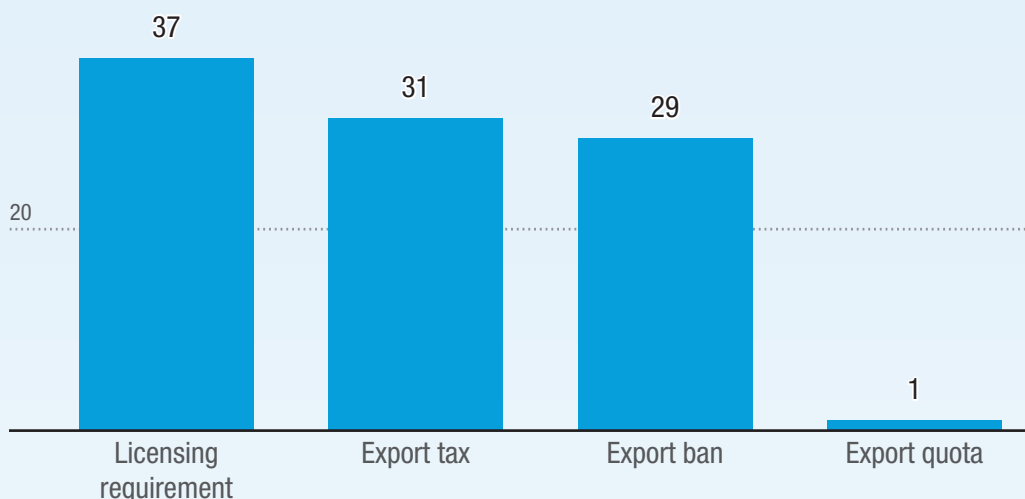
Licensing requirements and export taxes are among the most widely used measures (Figure 5). In recent years, however, there has been a notable shift toward more stringent instruments, including export bans and other quantitative restrictions. By contrast, export quotas are used less frequently.



Figure 5

Nearly 100 export measures on critical minerals have been introduced since 2020

Number of new export measures on critical energy transition minerals, January 2020 – May 2026



Source: UN Trade and Development (UNCTAD) based on UNCTAD research, Global Trade Alert and OECD Inventory of Export Restrictions on Critical Raw Materials 2026.

Since 2020, major global suppliers of critical minerals have introduced trade restrictive measures on CETMs. The Democratic Republic of the Congo has introduced the highest number of export measures (18), followed by China (16) and Indonesia (12). Burundi and the Bolivarian Republic of Venezuela follow, with 8 measures each, while Zimbabwe has introduced 7. These countries control large shares of global supply, so they could have greater market leverage and stronger incentives to manage exports strategically.

The data suggests that strategic motivations and policy instruments diverge sharply across implementing countries, ranging from security-oriented controls to revenue generation and the promotion of domestic processing. For example,

China's export measures consist of licensing requirements aimed at safeguarding national security and interests, protecting strategic resources, and fulfilling international non-proliferation obligations.⁷

Indonesia has adopted a mix of instruments, introducing five export bans and seven licensing requirements over the same period, aimed to support domestic processing and strengthen export monitoring.¹⁰

The Democratic Republic of the Congo relies predominantly on export taxes, which account for 12 of the measures it has introduced since 2020, followed by three export bans, two licensing requirements, and one export quota. These measures are aimed at different objectives such as generating revenue,⁸ creating incentives for domestic processing, and stabilizing cobalt prices amidst a multi-year low in 2025.⁹

The broader picture is clear: amid rising demand and the growing strategic importance of CETMs, governments with control over these materials are increasingly aware of their leverage and are actively integrating trade policy instruments into their toolkits to capture more domestic value by moving from raw material exports to processing and manufacturing, and by using export restrictions to promote national strategic objectives.

Major CETM importers are seeking to strengthen supply chain resilience

Concerns over concentration risks have grown as the supply of CETMs remains heavily dominated by a small number of countries, particularly in mining, processing, and refining activities. Amid rising demand for these materials, a number of major CETM importers, including the European Union, Japan and the United States, have adopted strategies to strengthen the resilience of their supply chains through diversification of import sources, domestic capacity development, recycling, and strategic partnerships.

⁷ <https://www.iea.org/policies/17893-announcement-on-the-implementation-of-export-control-of-items-related-to-gallium-and-germanium> and <https://www.iea.org/policies/26795-decision-to-implement-export-controls-on-tungsten-tellurium-bismuth-molybdenum-and-indium-related-items>

⁸ https://www.droitcongolais.info/files/610.12.22-Loi-du-28-decembre-2022_Exercice-2023.pdf and <https://www.iea.org/policies/28969-temporary-suspension-of-cobalt-export-from-the-democratic-republic-of-congo>

⁹ <https://www.spglobal.com/energy/en/news-research/latest-news/metals/022525-drc-attempts-to-address-surplus-cobalt-supply-declining-price-with-export-ban>

¹⁰ <https://setkab.go.id/en/govt-to-impose-export-ban-on-bauxite-ore-june-next-year/> and <https://jdih.kemendag.go.id/peraturan/peraturan-menteri-perdagangan-nomor-96-tahun-2019-tentang-ketentuan-ekspor-produk-pertambangan-hasil-pengolahan-dan-pemurnian>

The European Union has shaped recent policy through the Critical Raw Materials Act,¹¹ which sets quantitative targets for 2030 for domestic extraction, processing, and recycling of strategic minerals. The European Union has also strengthened external engagement through Strategic Partnerships, with 15 agreements signed to date, including with three least developed countries (LDCs). Complementary initiatives, such as Global Gateway,¹² aim to mobilize financing for infrastructure and processing projects in partner countries. In 2025, the European Investment Bank adopted a new initiative focused on CETMs, including an expected €2 billion in financing to support the objectives of the Critical Raw Material Act.¹³

Japan's CETM strategy is framed by its 2022 Economic Security Promotion Act¹⁴ and led by institutions such as the Japan Organization for Metals and Energy Security (JOGMEC). JOGMEC, for example, manages a programme that makes available subsidies to cover up to 50 per cent of the costs of eligible CETM projects across the value chain including mining and refining.¹⁵ There is also government support for stockpiling CETMs. Furthermore, Japan has signed CETM-related agreements with Angola, the Democratic Republic of the Congo and Namibia,¹⁶ reflecting an effort to diversify away from traditional suppliers.

The United States has adopted a comprehensive approach to CETM supply security, including equity participations, loans and incentives for the domestic mining and refining industry. A \$12 billion public-private stockpiling initiative, called Project Vault, was launched in 2026.¹⁷ Internationally, the United States has expanded its network of CETM agreements, signing 11 bilateral frameworks during a ministerial meeting in February 2026 that included the European Commission and 54 countries,¹⁸ five of them LDCs. Moreover, the United States proposed the establishment of a preferential trade zone for CETMs, which would include tariff-enforced price floors.¹⁹ In April 2026, the United States and the European Union further deepened cooperation through a Memorandum of Understanding (MoU) on a strategic partnership on CETMs.²⁰

Critical minerals partnerships are booming

UNCTAD research shows that since 2022, CETM partnership bilateral agreements have been rapidly expanding across the world, reflecting growing recognition of their importance.²¹ A small group of major economies is leading this expansion. The European Union, the United States, India, Canada, Indonesia, and Japan have entered into minerals agreements with a wide range of countries, generally showing limited overlap in their partner networks and suggesting a broad and competitive effort to secure access and influence across different regions. Among the 73 partnership agreements analysed by UNCTAD, 27 are between a developed and a developing country, 26 are between developed countries only, and 20 are between developing countries only (Figure 6).

¹¹ https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401252

¹² https://commission.europa.eu/topics/international-partnerships/global-gateway_en

¹³ <https://www.eib.org/en/press/all/2025-156-eib-steps-up-financing-for-european-security-and-defence-and-critical-raw-materials>

¹⁴ <https://www.wipo.int/wipolex/en/legislation/details/22642>

¹⁵ <https://www.iea.org/policies/24991-critical-minerals-subsidy-program>

¹⁶ <https://www.policycenter.ma/publications/japan-strengthen-to-africa>

¹⁷ <https://www.exim.gov/news/week-review-project-vault-and-strategic-critical-mineral-reserve>

¹⁸ <https://www.state.gov/releases/office-of-the-spokesperson/2026/02/2026-critical-minerals-ministerial>

¹⁹ <https://www.reuters.com/world/china/us-hosts-countries-talks-weaken-chinas-grip-critical-minerals-2026-02-04/>

²⁰ https://ec.europa.eu/commission/presscorner/detail/en/ip_26_862

²¹ Based on an analysis of the texts of 73 CETM international agreements and partnership instruments (e.g., MOUs, joint statements, and action plans) concluded since 2007, with 58 signed after 2022.

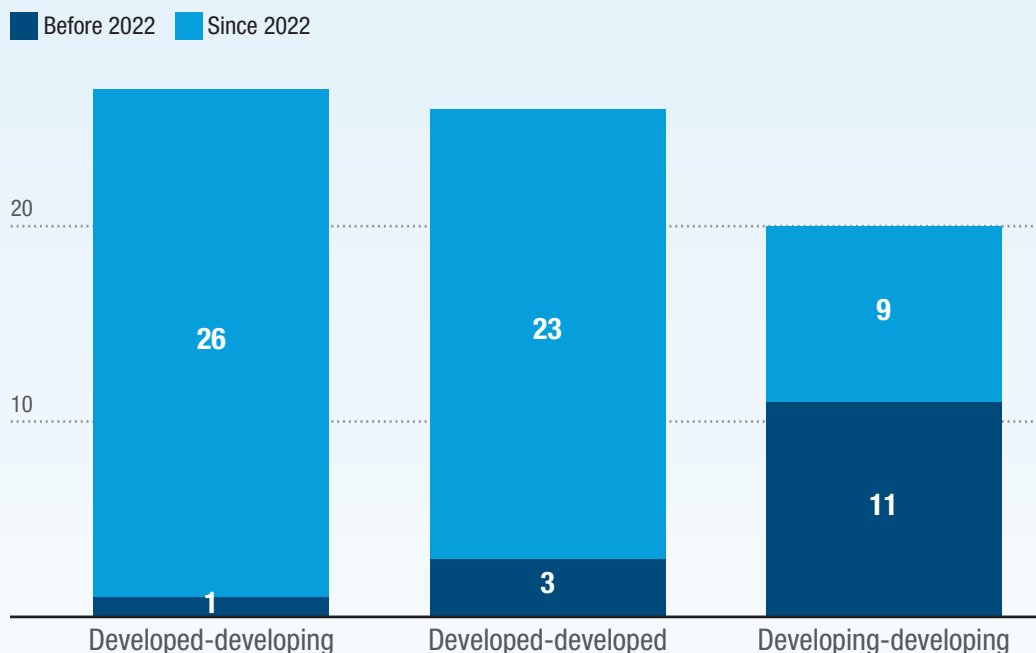




Figure 6

Critical mineral partnership agreements are proliferating

Number of critical mineral partnership agreements by development status of the partners



Source: UN Trade and Development (UNCTAD).

Note: Based on an analysis of 73 CETM international agreements and partnership instruments concluded since 2007.

Recent CETM partnership agreements are broadening their scope across the value chain. Many now aim to secure access to critical energy transition minerals from exploration and extraction through processing, refining, use, and recycling.

To achieve this, these arrangements combine trade, industrial, and investment policy tools. They often include trade-related provisions such as export measures and preferential arrangements, alongside industrial policy instruments, including subsidies, grants, and public financial support. In some cases, they also incorporate coordinated pricing mechanisms, offtake agreements, export credit support, and structured supply commitments, reflecting a more active role for governments in shaping markets and ensuring stable access.

Despite this breadth, upstream activities remain central. A defining feature of many agreements is their strong emphasis on extraction, underscoring the priority placed on securing raw material supply. However, patterns differ across country groupings. Agreements among advanced economies increasingly extend to midstream and downstream segments, such as processing, refining, and manufacturing, while those involving developing countries tend to focus more narrowly on upstream cooperation, with fewer provisions supporting value addition. Notable exceptions are emerging, such as the United States–Democratic Republic of the Congo–Zambia initiative, which promotes electric vehicle battery value chains and creates opportunities for downstream industrial development.



The design of these agreements also reflects evolving policy approaches. For example, United States-led partnerships emphasize building secure and diversified supply chains among partners by stabilizing markets, coordinating policies, and reducing reliance on dominant suppliers. They include a range of trade-related instruments, such as preferential trade elements, “fair” or “market-oriented” conditions, and in some cases, emerging mechanisms such as price coordination frameworks or price floors.

These instruments often go beyond traditional trade measures. Rather than focusing solely on market access, many operate through hybrid arrangements that combine trade and industrial policy features. As a result, their consistency with WTO disciplines, particularly those relating to tariffs, quotas, subsidies, and non-discrimination, remains context-specific and, in many cases, not yet fully tested. Moreover, many provisions are still at a framework or coordination stage, with their detailed design and implementation remaining under development, leaving their ultimate impact uncertain.

These dynamics raise important development and systemic considerations. Critical mineral-rich developing countries, often negotiating individually, risk becoming locked into roles centered on raw material extraction, limiting their ability to capture higher-value segments of the chain. To address this, they may seek stronger provisions on local processing, technology transfer, and skills development to support more diversified and sustainable participation.

Ensuring coherence with the multilateral trading system will be essential. Maintaining transparency, predictability, and non-discrimination can help prevent fragmentation into competing blocs and support the emergence of CETM markets that are both resilient and inclusive, while effectively contributing to the energy transition.





Critical minerals trade policy at a crossroads: towards fragmentation or cooperation?

The trade policy environment surrounding CETMs is undergoing a significant transformation. Both exporting and importing countries are increasingly deploying trade policy as a strategic instrument to achieve their broader economic, industrial, and geopolitical objectives. The growing use of international CETM partnership agreements between producer and consumer countries is reshaping market dynamics, often prioritizing supply security, resilience, and strategic alignment over cost efficiency.

At the same time, the rising demand for CETMs is intensifying competition for access to these resources. In this context, trade policy is playing a more prominent role in shaping where, how, at what cost, and for whose benefit CETMs are produced, processed, and traded across the global economy.

For developing countries endowed with significant CETM resources, this evolving landscape presents both opportunities and risks. On the one hand, robust global demand offers the potential for increased export revenues, greater foreign direct investment inflows, and opportunities to develop downstream processing and manufacturing activities that can support structural transformation. On the other hand, there remains a risk of reinforcing traditional extractive patterns, whereby countries continue to export raw materials with limited domestic value addition.

While well-intentioned, such agreements also carry the risk of contributing to an increasingly fragmented regulatory landscape. The proliferation of distinct rules, standards, and eligibility criteria across agreements may result in a “spaghetti bowl” of overlapping and sometimes inconsistent requirements. This could inadvertently raise transaction costs for firms, complicate compliance, and create uncertainty for investors. In some cases, divergent provisions may even generate competing or mutually exclusive conditions for participation in supply chains, ultimately undermining the efficiency and resilience that these initiatives seek to promote.

As competition for CETMs intensifies, critical-mineral-rich developing countries may increasingly be pressured to align with one importing partner at the expense of others. Such forced choices could not only weaken the bargaining position of individual exporters but also fragment global CETM trade into competing blocs, raising costs and reducing efficiency for all participants. A more coordinated, multilateral approach could help avoid this outcome, preserving developing countries' ability to engage with multiple partners and ensuring that critical mineral trade remains open, competitive, and conducive both to meeting the development needs of critical-mineral-rich developing countries and to delivering an affordable, timely energy transition for all.



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